

exercise 03: SQL Foundations

01

```
SELECT Product-name,  
       Price,  
       CASE
```

```
       WHEN price > 1000 THEN 'Expensive'
```

```
       WHEN price BETWEEN 100 AND 1000  
       THEN 'mid-range'
```

```
       ELSE 'Budget'
```

```
END AS price-category
```

```
FROM products;
```

Product-name	price	price-category
Laptop	1200.00	Expensive
Phone	800.00	mid-range
Keyboard	45.00	Budget
Monitor	300.00	mid-range
Mouse	25.00	Budget

02

```
SELECT customer-name,  
       Amount,  
       CASE
```

```
       WHEN value ≥ 1000 THEN 'High value'
```

```
       WHEN value BETWEEN 500 AND 999.99  
       THEN 'medium value'
```

```
       ELSE 'Low value'
```

```
END AS order-value category
```

```
FROM orders;
```

Customer-name	amount	order-value category
Alice	150.00	Low value
Bob	560.00	medium value
Charlie	999.99	medium value
Diana	45.50	Low value
Ethan	1200.00	High value

03

```

SELECT emp-name,
       department,
       Salary,
CASE
    WHEN department = 'IT' AND AND Salary
        ≥ 20000 THEN 'Senior IT'
    WHEN department = 'HR' AND Salary
        > 55000 THEN 'Experience HR'
    ELSE 'Staff'
END AS position-level

```

FROM employees;

emp-name	department	Salary	Position-level
John	IT	28000	Senior IT
Sara	HR	60000	Experience HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

04

```

SELECT student-name,
       Score,
CASE
    WHEN Score ≥ 90 THEN 'A'
    WHEN Score BETWEEN 80 AND 89
        THEN 'B'

```

WHEN score BETWEEN 70 AND 79
THEN 'C'

WHEN score BETWEEN 60 AND 69
THEN 'D'

ELSE 'F'

END AS grade

FROM students;

Student-name	Score	grade
Anna	92	A
Ben	76	C
Carla	59	F
David	83	B
Ella	68	D

Q5

SELECT delivery-id,
delivery-time-minute^(d-t-m),
CASE

WHEN d-t-m ≤ 30 THEN 'Fast'

WHEN d-t-m BETWEEN 31 AND 60 THEN
'On Time'

ELSE 'Late'

END AS performance

FROM deliveries;

delivery-id	delivery-time-minute	performance
1	45	on-time
2	60	late
3	30	Fast
4	65	late
5	100	late

06

```
SELECT issue-type,  
       priority,  
       CASE  
         WHEN priority = 1 THEN 'low'  
         WHEN priority = 2 THEN 'medium'  
         ELSE 'High'  
       END AS priority-label  
FROM tickets;
```

issue-type	priority	priority-label
Login issue	1	low
Server down	3	High
Slow system	2	medium
Email error	2	medium
password reset	1	low

07

```
SELECT student-id,  
       total days,  
       CASE  
         WHEN attendance percentage  $\geq 90\%$  THEN  
           'Excellent'  
         WHEN attendance percentage BETWEEN 75%  
           AND 89% THEN 'Good'  
         ELSE 'needs improvement'  
       END AS attendance status
```

```
FROM attendance;
```

student-id	attendance percentage	attendance-status
1	50	Needs improvement
2	50	Needs improvement
3	50	Needs improvement
4	50	Needs improvement
5	50	Needs improvement

08

```
SELECT product-id,  
       stock-qty,  
       CASE
```

```
       WHEN stock-qty = 0 THEN 'out of stock'
```

```
       WHEN stock-qty BETWEEN 1 AND 5 THEN  
       'low stock'
```

```
       ELSE 'in stock'
```

```
END stock-status
```

```
FROM products-inventory;
```

product-id	stock-qty	stock-status
1	5	in stock
2	0	out of stock
3	25	in stock
4	10	in stock
5	3	low stock

9

```
SELECT subject,  
       enrolled_students,  
       CASE
```

```
       WHEN enrolled_students ≥ 25 THEN 'large'
```

```
       WHEN enrolled_students BETWEEN 10 AND  
       24 THEN medium
```

```
       ELSE 'small'
```

```
END class-size category
```

```
FROM classes;
```

subject	enrolled_students	class-size category
math	30	large
English	25	large
science	15	medium
Art	5	small
History	20	medium

10

```

SELECT payment-id
       payment-method
       amount,
       CASE
         WHEN payment-method = 'CASH' AND
              amount ≥ 200 THEN 'eligible'
         ELSE 'not eligible'
       END AS discount-eligibility
FROM payments;

```

Payment-id	payment-method	amount	discount-eligible
1	card	50.00	not eligible
2	CASH	200.00	Eligible for discount
3	card	150.00	not eligible
4	Paypal	75.00	not eligible
5	CASH	300.00	Eligible for discount